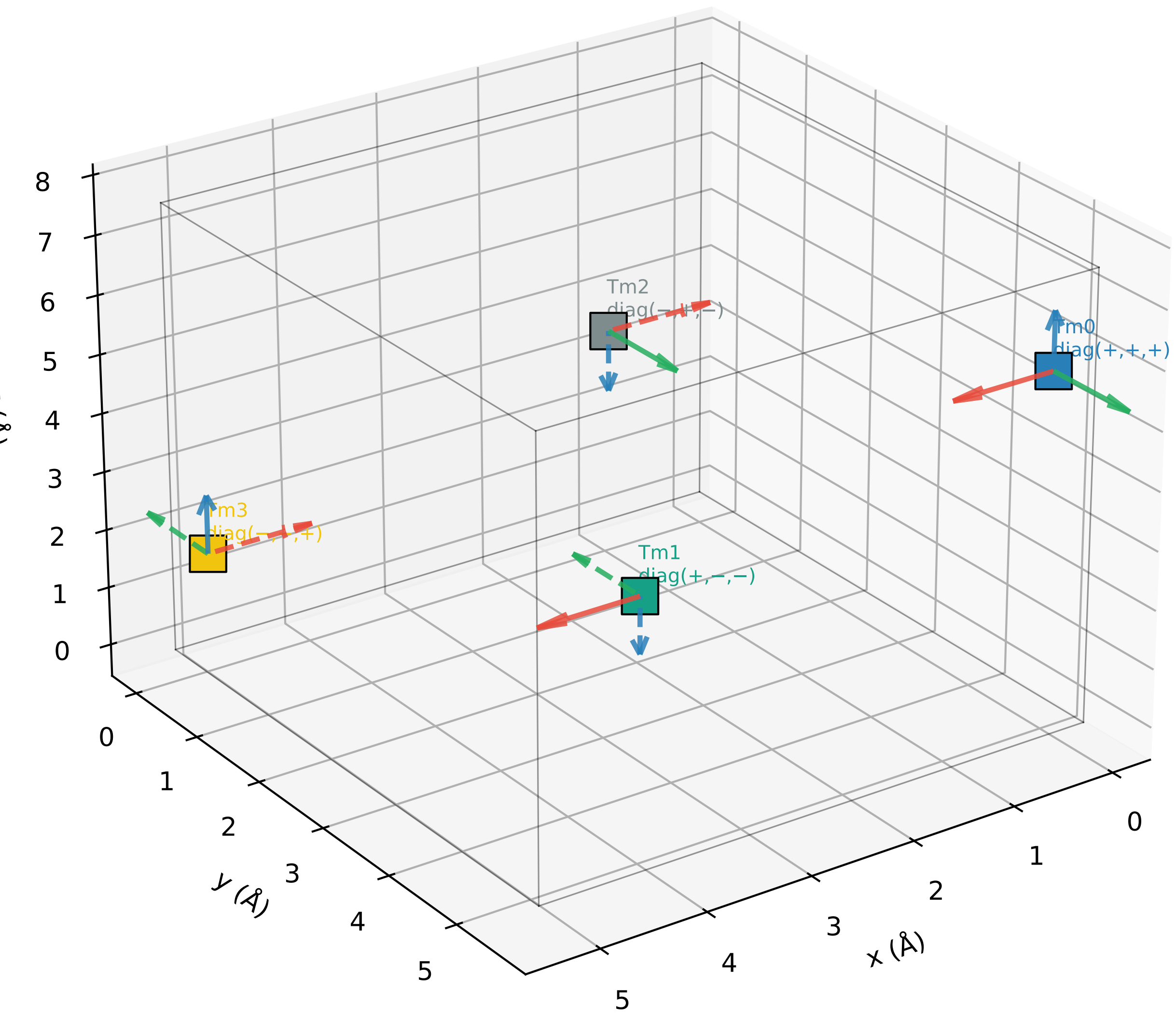
Tm sublattice local frames (tmfeo3_tm_local_frames_xyz)
The η^z_Tm pattern is the detector sensitivity for H//z (c-axis) spectroscopy

Tm local frame arrows at each sublattice position



Component	sub 0	sub 1	sub 2	sub 3	Pattern
η^x_Tm (R_xx)	+1	+1	−1	−1	σ_A
η^y_Tm (R_yy)	+1	−1	+1	−1	σ_G
η^z_Tm (R_zz)	+1	−1	−1	+1	σ_C

```
Detector coupling for H//z (c-axis field / detection):

  M_z^Tm = Σ_j η^z_Tm,j · c_2 · λ²_j
         = σ_C · λ²_vector

If λ² has sigma pattern σ_P, then:
  M_z^Tm = σ_C · σ_P · λ²_avg

Nonzero only if σ_C · σ_P = 4, i.e. σ_P = σ_C

chi2z drives λ² with σ_F pattern:
   σ_C · σ_F = 0  →  M_z^Tm = 0  (FORBIDDEN)

chi2x drives λ² with σ_A pattern (if F_x ≠ 0):
   σ_C · σ_A = 0  →  M_z^Tm = 0  (FORBIDDEN)

For M_z^Tm to be nonzero, λ² must carry σ_C pattern.
chi2y would drive λ² with σ_C pattern (η^y_Fe = σ_C),
because σ_C · σ_G ≠ 0... but wait: in Γ₂, S^y_Fe = 0.
So even chi2y cannot generate λ² in the equilibrium state.
```